

10_Database_Architecture.md

Part 2 — Schema Design & Relationships

Database Schema Overview

``text id="cv8d0m" Users | |— Portfolios | |— Holdings | |— Transactions | |—
Portfolio Analytics | |— Watchlists | |— AI Conversations | |— Learning Progress | |—
Competition Participants | |— Notifications

Stocks | |— Stock Prices |— Historical Prices |— Stock Categories

```
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# Table: users

## Purpose

Stores registered user information.

### Primary Fields

``text id="hxyclm"
id

clerk_user_id

full_name

email

avatar_url

role

created_at

updated_at
```

Table: portfolios

Purpose

Stores each user's virtual portfolio.

Fields

```text id="mcn18k" id

user\_id

virtual\_balance

portfolio\_value

total\_profit\_loss

created\_at

updated\_at

Relationship

```text id="hvhhv8"

User

1 → 1

Portfolio

Table: holdings

Purpose

Stores active stock holdings.

Fields

```text id="axghwt" id

portfolio\_id

stock\_id

quantity

average\_buy\_price

current\_price

current\_value

profit\_loss

updated\_at

Relationship

```text id="zjlm8"

Portfolio

1 → N

Holdings

Table: transactions

Purpose

Stores complete trading history.

Fields

```text id="4i9fdm" id

portfolio\_id

stock\_id

transaction\_type

quantity

price

total\_amount

executed\_at

### Transaction Types

- BUY
- SELL

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# Table: stocks

## Purpose

Stores listed companies.

### Fields

```text id="8lmk3l" id

id

symbol

company_name

sector

industry

exchange

is_active

Table: stock_prices

Purpose

Stores latest market prices.

Fields

```text id="ccu7ol" id

stock\_id

current\_price

day\_high

day\_low

volume

market\_cap

updated\_at

Relationship

```text id="0f8n8m"

Stock

1 → 1

Stock Price

Table: watchlists

Purpose

Stores bookmarked stocks.

Fields

```text id="dyyz9z" id

user\_id

stock\_id

created\_at

Relationship

```text id="crgkg5"

User

1 → N

Watchlist

Table: ai_conversations

Purpose

Stores AI interaction history.

Fields

``text id="jlwmzj" id

user_id

title

created_at

updated_at

Table: ai_messages

Purpose

Stores conversation messages.

Fields

``text id="3drw5r"

id

conversation_id

role

content

created_at

Roles

- USER
- ASSISTANT
- SYSTEM

Table: learning_progress

Purpose

Tracks learning progress.

Fields

``text id="4s0nhf" id

user_id

course_id

lesson_id

completion_percentage

completed_at

```
---

# Table: competitions

## Purpose

Stores competition information.

### Fields

```text id="xgk5ks"
id

title

description

start_date

end_date

status

created_at
```

# Table: competition\_participants

## Purpose

Stores competition participants.

## Fields

``text id="rtw0zh" id

competition\_id

user\_id

rank

portfolio\_value

profit\_loss

```

Table: notifications

Purpose

Stores system notifications.

Fields

``text id="j7f0mh"
id

user_id

title

message

type

is_read

created_at
```



## Relationship Summary

```text id="fw7m9p" User | | Portfolio (1:1) | | Holdings (1:N) | | Transactions (1:N) |  
| | Watchlist (1:N) | | AI Conversations (1:N) | | Learning Progress (1:N) | | Notifications
(1:N) | | Competition Participants (1:N)```

Stock | | Stock Price (1:1) | | Holdings (1:N) | | Transactions (1:N) | | Watchlist (1:N)```

Constraints

- Email must be unique.
 - Clerk User ID must be unique.
 - Stock symbol must be unique.
 - Foreign keys required for all relationships.
 - Cascade delete only where appropriate.
 - Prevent orphan records.
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Indexing Strategy

Create indexes for:

- email
 - clerk_user_id
 - stock_symbol
 - user_id
 - portfolio_id
 - competition_id
 - created_at
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Validation Checklist

- Normalized schema
 - Strong relationships
 - Minimal duplication
 - Indexed search fields
 - Consistent naming
 - Future scalability
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Next Document

10_Database_Architecture.md — Part 3